



7CS039 – Statistics for Data Science

School of Mathematics and Computer Science

Faculty of Science and Engineering

Insights into Malignant Melanoma Outcomes: An Exploratory Data Analysis of Key Factors and Trends

Presented by:

Solagbade Akande

2417206

16th January, 2025

Table of Contents

1. Introduction	3
2. Data Overview and Methodology	3
3. Summary Statistics.....	3
4. Exploring Survival Trends Through Graphical Summaries	4
5. Regression and Correlation	6
6. Gender-Based Outcome Analysis: Two-Sample Significance Tests	9
6.1 Chi-Square Test for the Relationship Between Survival Time and Gender.....	9
6.2 t-Test for the Relationship Between Age and Gender.....	9
6.3 Mann-Whitney U Test for the Relationship Between Tumor Thickness and Gender.....	10
7. Analyzing Gender-Based Distributions with QQ-Plots.....	11
8. Key Insights and Recommendations	12
References.....	13

1. Introduction

This report analyzes malignant melanoma patient survival outcomes using statistical methods and visualizations in R. It identifies key factors influencing survival, following principles and techniques from the module's lecture notes (Rodden, 2024).

This report, created in Quarto using RStudio, was converted to Word and finalized as a PDF. Key code snippets and charts, presented as embedded code or screenshots, were retained throughout.

2. Data Overview and Methodology

The dataset contains survival data for malignant melanoma patients treated at the University Hospital of Odense, Denmark (1962–1977). Key variables include survival time, tumor thickness, ulceration status, and demographics such as age and sex (University Hospital of Odense, Denmark, 1977). The data cleaning process involves renaming columns for clarity, re-coding variables with descriptive labels, and checking for missing values to ensure the dataset is well-organized for analysis.

```
{r}
melanoma <- read.csv("melanoma.csv") # Load and clean dataset
head(melanoma) # Display dataset
# Recode Variables
melanoma$sex <- factor(melanoma$sex, levels = c(0, 1), labels = c("Female", "Male"))
melanoma$ulcer <- factor(melanoma$ulcer, levels = c(0, 1), labels = c("Absent", "Present"))
melanoma$status <- factor(melanoma$status, levels = c(1, 2, 3), labels = c("Melanoma Death", "Alive", "Unrelated Death"))
```

3. Summary Statistics: Overall summary of the Dataset for Quick Reference

Descriptive statistics form the foundation for practically all statistical analysis (Donnelly & Abdel-Raouf, 2016)

```
{r}
summary_stats <- summary(melanoma)
# Convert the summary statistics into a kable table
kable(summary_stats, caption = "Overall Summary of the Melanoma Dataset")
```

Overall Summary of the Melanoma Dataset						
time	status	sex	age	year	thickness	ulcer
Min.: 10	Melanoma Death: 57	Female: 126	Min.: 4.00	Min.: 1962	Min.: 0.10	Absent: 115
1st Qu.: 1525	Alive: 134	Male: 79	1st Qu.: 42.00	1st Qu.: 1968	1st Qu.: 0.97	Present: 90
Median: 2005	Unrelated Death: 14	NA	Median: 54.00	Median: 1970	Median: 1.94	NA
Mean: 2153	NA	NA	Mean: 52.46	Mean: 1970	Mean: 2.92	NA
3rd Qu.: 3042	NA	NA	3rd Qu.: 65.00	3rd Qu.: 1972	3rd Qu.: 3.56	NA
Max.: 5565	NA	NA	Max.: 95.00	Max.: 1977	Max.: 17.42	NA

The dataset reveals key insights into melanoma patients and their treatment journeys:

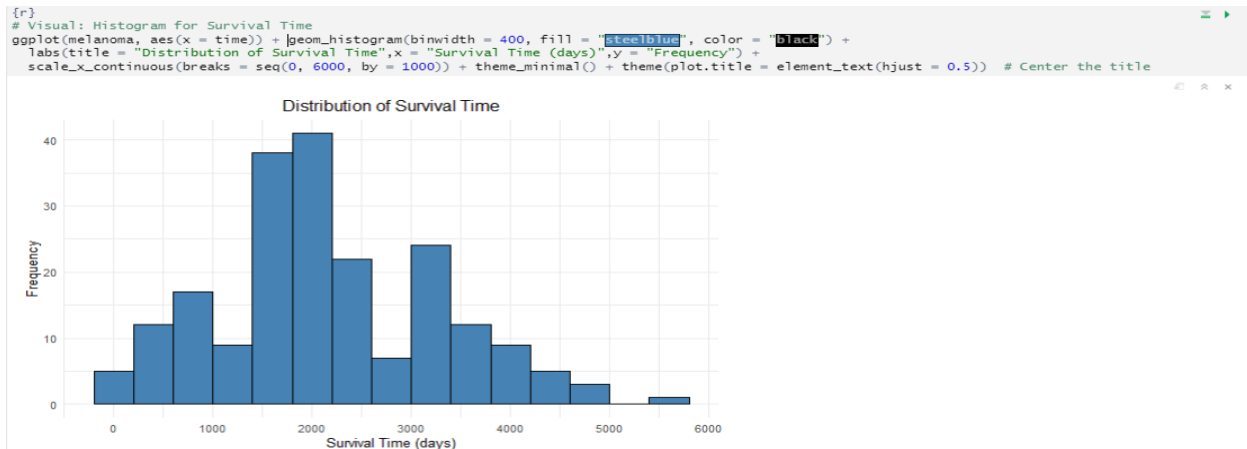
- **Survival Time:** Recorded survival times range from **10 days** to approximately **15 years** (calculated as **5565÷365**). The average survival time is **5.87 years (mean of 2135 days)**, with a median survival time of **5.49 years (2005 days)**. These figures highlight the variability in patient outcomes and raise questions about the factors influencing survival.
- **Patient Status:** Of the patients, **57 (27.8%)** died from melanoma, **134 (65.4%)** survived, and **14 (6.8%)** passed away due to unrelated causes. These figures reflect the significant impact of the disease and the complexity of health outcomes.
- **Sex:** Women make up **61.5%** of the dataset (**126 patients**), while men account for **38.5% (79 patients)**. This suggests trends may lean toward female experiences, highlighting the importance of ensuring male-specific insights are not overlooked.
- **Age:** The ages of patients range from **4 to 95 years**, with a **median age of 54 years and an average of 52.46 years**. This demographic diversity spans multiple life stages, offering a broad view of melanoma's impact across different age groups.
- **Year of Operation:** The operations were performed between **1962 and 1977**, with a median year of **1970**. This consistent timeline allows for reliable analysis of treatment trends during this period.
- **Tumor Thickness:** Tumor thickness varies significantly, ranging from **0.1 mm to 17.42 mm**, with a **median of 1.94 mm and an average of 2.92 mm**. This variability underscores the differing levels of disease severity faced by patients.
- **Ulceration:** Ulceration is absent in **56% of cases (115 patients)** and present in **44% of cases (90 patients)**. Its association with survival outcomes warrants further investigation.

These insights provide a foundation for exploring the factors that drive outcomes, helping us better understand the experiences of melanoma patients.

4. Exploring Survival Trends Through Graphical Summaries

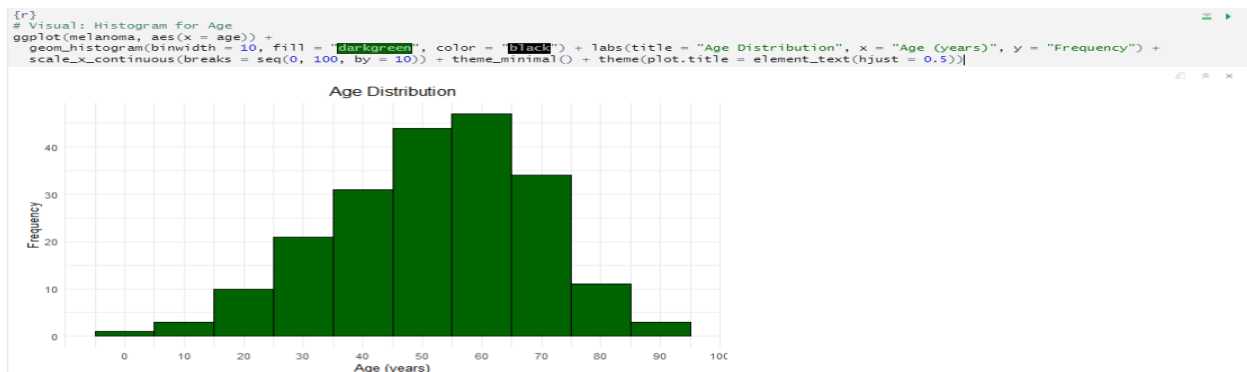
Graphical summaries illustrate survival trends across key variables in the dataset.

4.1 Survival Time ('time')



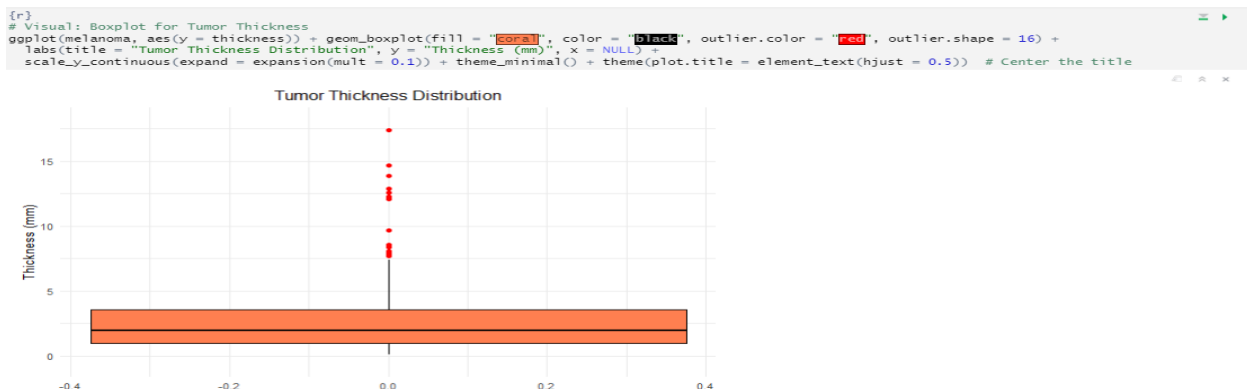
The histogram shows most patients survived between **1000 and 3000 days**, peaking around **2000 days (5.5 years)**. Long-term survival beyond **3000 days** was less common, with only a few reaching **10 years**, underscoring the need to explore factors influencing extended survival.

4.2 Age ('age')



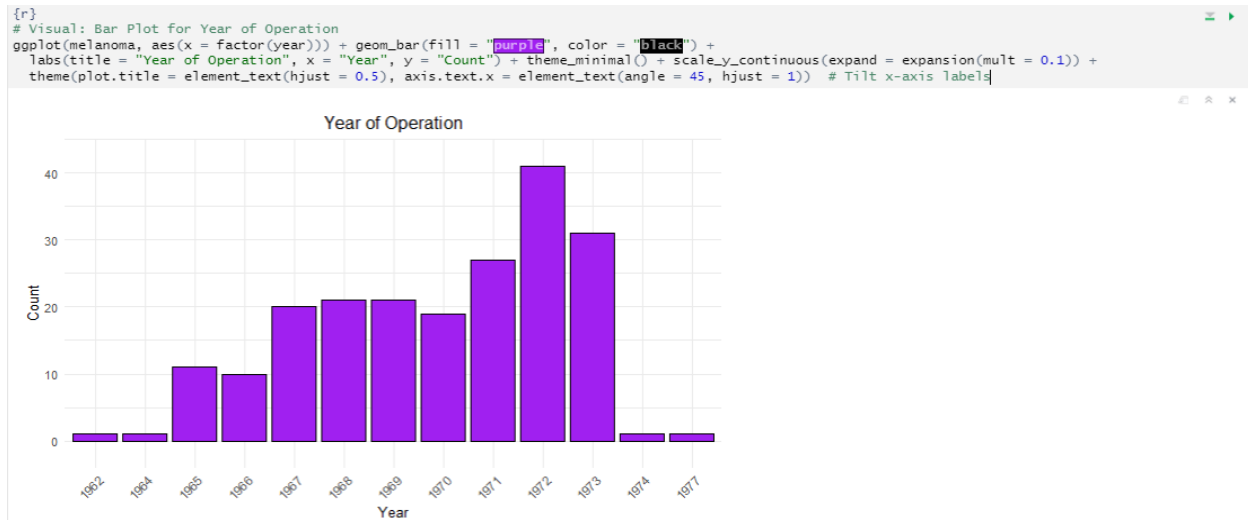
The histogram shows melanoma patients range from **under 10 to over 90 years**, peaking between **50-60 years**, indicating it's most common in older adults but can affect younger individuals. This insight aids targeted awareness and prevention.

4.3 Tumor Thickness



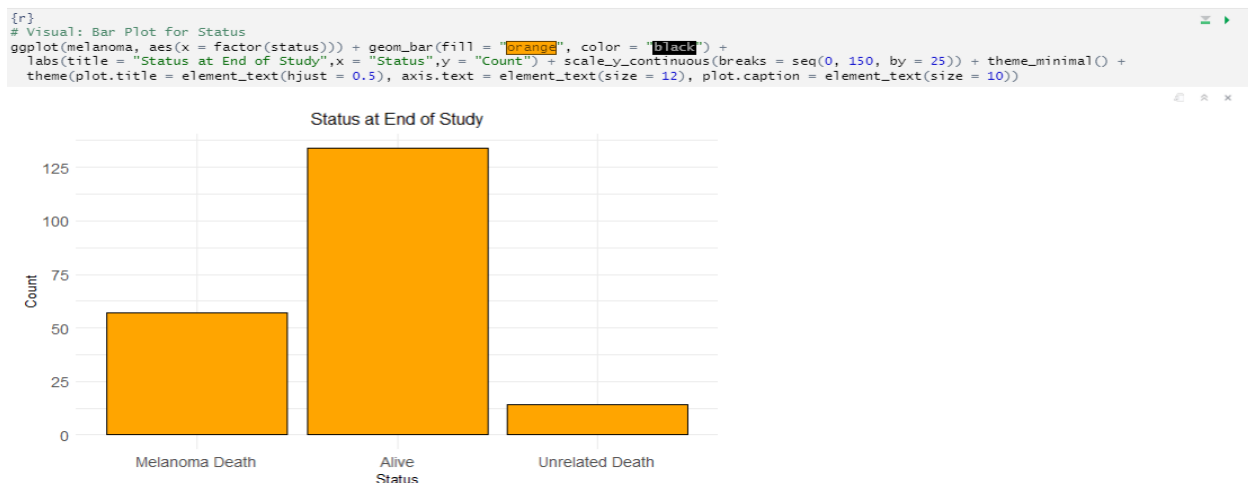
The boxplot above shows that most tumors are **under 5 mm thick**, with a median of approximately **2.9 mm**. Outliers, representing tumor thicknesses exceeding **10 mm**, highlight more severe cases likely linked to worse survival outcomes. This variability underscores the importance of understanding tumor thickness's role in patient prognosis.

4.4 Year of Operation



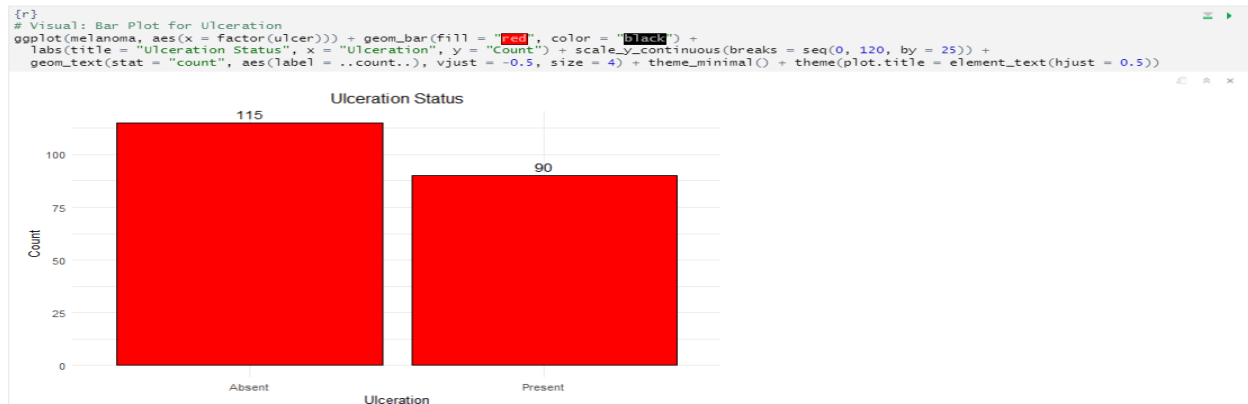
The bar plot above reveals consistent melanoma surgeries from **1962 to 1977**, peaking significantly in **1972**. This peak may reflect increased awareness, policy changes, or improved detection methods. A sharp **decline after 1972**, following stable numbers from **1967 to 1971**, could indicate reduced data collection or shifts in treatment strategies.

4.5 Status



The bar plot shows most patients in the **"Alive Status"** category, indicating positive survival outcomes. Smaller groups include **"Melanoma Death"**, highlighting the disease's impact, and **"Unrelated Death"**, reflecting other factors. This distribution emphasizes melanoma's severity while showcasing the potential for survival with effective treatment.

4.6 Ulceration



The bar plot above shows that **115 patients (54%)** had no ulceration, while **90 (46%)** had ulcerated tumors, indicating ulceration is a common feature in melanoma cases. This near-even distribution highlights the importance of analyzing its impact on survival outcomes and treatment effectiveness.

4.7 Sex



The bar plot reveals that the dataset consists of **126 females (61.5%)** and **79 males (38.5%)**, indicating a gender imbalance with more females treated for melanoma. This distribution may affect trends in survival outcomes and other variables. Analyzing gender-specific patterns could uncover differences in melanoma progression or treatment effectiveness.

5. Regression and Correlation: Understanding Key Relationships (Analysis and Commentary)

To uncover meaningful insights, we use regression analysis and correlation to explore how **survival time**, **tumor thickness**, and **age** interact. These statistical methods help quantify **relationships** and reveal **patterns** that inform our understanding of patient outcomes.

Step 1: Calculating the Correlation Coefficient (r): An **r** value near **1** indicates a **strong positive relationship**, near **-1** a **strong negative relationship**, and near **0** no meaningful linear connection.

```
[r]
# Correlation
cor_thickness_time <- cor(melanoma$time, melanoma$thickness, method = "pearson")
cat("Correlation (time ~ thickness):", cor_thickness_time, "\n")

cor_age_time <- cor(melanoma$time, melanoma$age, method = "pearson")
cat("Correlation (time ~ age):", cor_age_time, "\n")

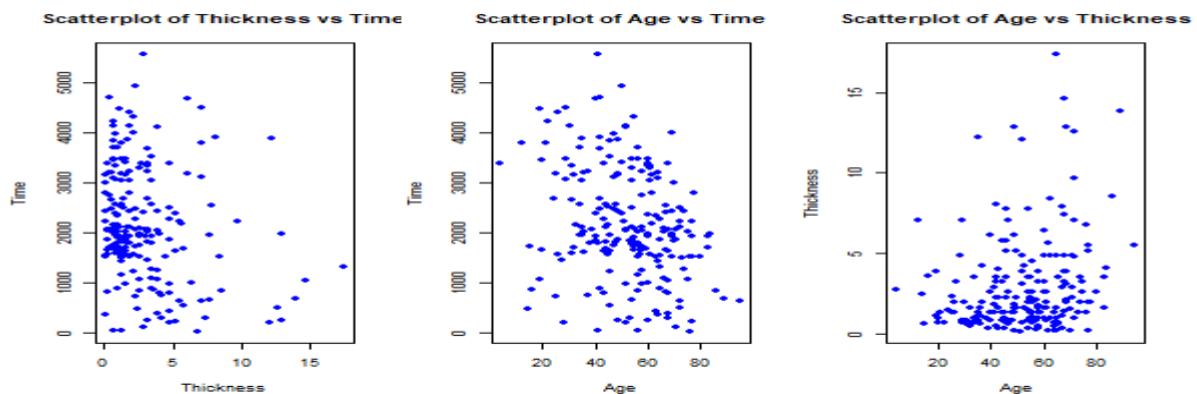
cor_age_thickness <- cor(melanoma$thickness, melanoma$age, method = "pearson")
cat("Correlation (thickness ~ age):", cor_age_thickness, "\n")
```

Correlation (time ~ thickness): -0.2354087
Correlation (time ~ age): -0.3015179
Correlation (thickness ~ age): 0.2124798

The correlation coefficients indicate **weak** relationships: $r=-0.235$ (time ~ thickness) shows a **slight** decrease in survival time as tumor thickness increases, $r=-0.302$ (time ~ age) suggests a **slight** decrease in survival time with increasing age, and $r=0.212$ (thickness ~ age) indicates a **slight** increase in tumor thickness with age.

Step 2: Visualizing the Data

```
{r}
# Set up a 1x3 grid for the plots
par(mfrow = c(1, 3))
# Scatterplot of Thickness vs Time, Age vs Time, Age vs Thickness
plot(melanoma$thickness, melanoma$time, main = "Scatterplot of Thickness vs Time",
     xlab = "Thickness", ylab = "Time", pch = 19, col = "blue")
plot(melanoma$age, melanoma$time, main = "Scatterplot of Age vs Time",
     xlab = "Age", ylab = "Time", pch = 19, col = "blue")
plot(melanoma$age, melanoma$thickness, main = "Scatterplot of Age vs Thickness",
     xlab = "Age", ylab = "Thickness", pch = 19, col = "blue")
```



The scatterplots (time ~ thickness, time ~ age, thickness ~ age) show no clear or strong relationships, with points widely scattered in each case, indicating minimal direct associations between these variables.

Step 3: Fitting a Linear Regression Model

Fitting linear regression models helps summarize the relationships between variables using simple equations:

- **time ~ thickness:** $y = mx + c$ where **y** represents **survival time**, **x** is **tumor thickness**, **m** is the **slope** showing how thickness impacts time, and **c** is the **intercept**.
- **time ~ age:** $y = mx + c$, where **y** is **survival time**, **x** is **age**, and **m** indicates the effect of age on time.
- **thickness ~ age:** $y = mx + c$, where **y** is **tumor thickness**, **x** is **age**, and **m** shows how age impacts thickness.

```
{r}
# Function to extract slope, intercept, and R-squared
extract_model_info <- function(model) {
  intercept <- coef(model)[1]
  slope <- coef(model)[2]
  r_squared <- summary(model)$r.squared
  return(c(Intercept = intercept, Slope = slope, R_Squared = r_squared))
}

# Fit and extract details for Time ~ Thickness
model_thickness <- lm(formula = time ~ thickness, data = melanoma)
cat("Model: Time ~ Thickness\n")
print(extract_model_info(model_thickness))

# Fit and extract details for Time ~ Age
model_age_time <- lm(formula = time ~ age, data = melanoma)
cat("\nModel: Time ~ Age\n")
print(extract_model_info(model_age_time))

# Fit and extract details for Thickness ~ Age
model_age_thickness <- lm(formula = thickness ~ age, data = melanoma)
cat("\nModel: Thickness ~ Age\n")
print(extract_model_info(model_age_thickness))
```

Model: Time ~ Thickness		
Intercept. (Intercept)	Slope.thickness	R_Squared
2413.41019325	-89.25453935	0.05541724
Model: Time ~ Age		
Intercept. (Intercept)	Slope.age	R_Squared
3217.44756962	-20.29314289	0.09091307
Model: Thickness ~ Age		
Intercept. (Intercept)	Slope.age	R_Squared
0.94105107	0.03771776	0.04514766

Model Summaries:

1. *Time ~ Thickness*

- **Equation:** $\text{Time} = -89.25 \times \text{Thickness} + 2413.41$
- **Slope:** Survival time **decreases** by **89.25 units** for every unit increase in **thickness**.
- **Intercept:** Predicted survival time is **2413.41** days (6.6 years) when thickness is zero.
- **R-squared:** **0.0554** (5.54% of survival time variability is explained by thickness), indicating a **weak relationship**, consistent with the scatterplot.

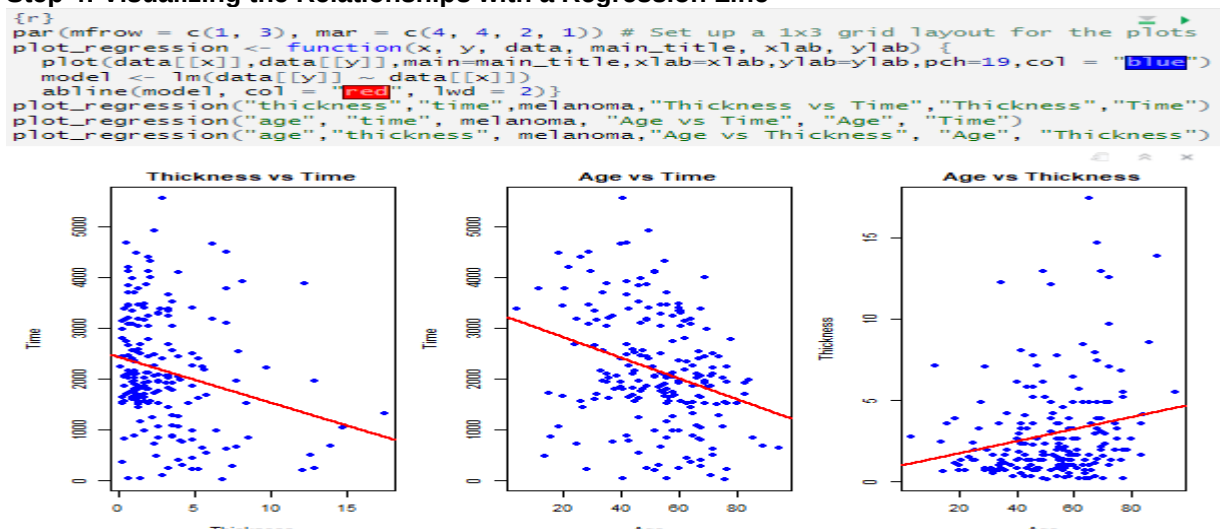
2. *Time ~ Age*

- **Equation:** $\text{Time} = -20.29 \times \text{Age} + 3217.45$
- **Slope:** Survival time **decreases** by **20.29 units** for each additional year of **age**.
- **Intercept:** Predicted survival time is **3217.45** days (8.8 years) when age is zero.
- **R-squared:** **0.091** (9.1% of survival time variability is explained by age), confirming a **weak relationship**.

3. *Thickness ~ Age*

- **Equation:** $\text{Thickness} = 0.0377 \times \text{Age} + 0.941$
- **Slope:** Tumor thickness **increases** by **0.0377 mm** per additional year of **age**.
- **Intercept:** Predicted tumor thickness is **0.941 mm** when age is zero.
- **R-squared:** **0.045** (4.5% of thickness variability is explained by age), reflecting a **weak relationship**.

Step 4: Visualizing the Relationships with a Regression Line



Time ~ Thickness: The scatterplot with a regression line shows a weak negative trend, indicating tumor thickness is not a strong predictor of survival. The variability in points suggests that additional factors are needed to explain survival outcomes.

Time ~ Age: The regression line reveals a slight downward trend, indicating a weak negative relationship where survival time decreases slightly with age. However, the minimal slope and variability in points highlight that age alone is not a strong predictor of survival outcomes, underscoring the need to explore other factors.

Thickness ~ Age: The scatterplot with a regression line shows a slight upward trend, suggesting a weak positive relationship between age and tumor thickness. However, the considerable variability in points indicates limited predictive value of age for tumor thickness.

Final Insights: Through correlation analysis, scatterplots, and regression modeling, we found that tumor thickness, age, and survival time all exhibit weak relationships. These findings highlight the need to examine additional variables to better understand factors influencing survival outcomes.

6. Gender-Based Outcome Analysis: Two-Sample Significance Tests for Key Variables

We will analyze survival time, tumor thickness, and age differences between genders using two-sample statistical tests to identify meaningful patterns.

Objective

Determine whether differences in key variables between men and women are statistically significant or due to chance.

Statistical Testing Framework:

- **Purpose:** Assess gender-based differences in **survival time, tumor thickness, and age**.
- **Key Metric:**
 - **p-value:** This shows how likely it is that the results happened by chance. If the p-value is less than 0.05, it means the results are likely meaningful and not just random.
- **Significance Level (α):** Set at 0.05, allowing a 5% margin of error.

Hypotheses and Decision Rule:

- **Null Hypothesis (H_0):** No significant difference between genders.
- **Alternate Hypothesis (H_1):** Significant difference exists.
- **Decision Rule:**
 - If $p < 0.05$, reject H_0 ; there is a significant difference.
 - If $p \geq 0.05$, do not reject H_0 ; insufficient evidence of a difference.

6.1 Chi-Square Test for the Relationship Between Survival Time and Gender

To determine if survival time differs by gender, we categorize survival into **Short**, **Medium**, and **Long** bins based on quartiles. Since survival time isn't normally distributed, the Chi-Square test is appropriate for this categorical analysis.

Hypotheses:

- **Null Hypothesis (H_0):** Survival time is independent of gender.
- **Alternate Hypothesis (H_1):** Survival time depends on gender.

Method: Survival time is binned into quartiles to create categories. A contingency table is generated to show the distribution of survival categories by gender, and a Chi-Square test is conducted using the **summary** function to assess significance.

```
{r}
# Categorize Survival Time (Short Survival, Medium Survival, Long Survival)
melanoma$time_bin <- cut(melanoma$time, breaks = quantile(melanoma$time, probs = c(0, 0.25, 0.75, 1), na.rm = TRUE),
  labels = c("Short Survival", "Medium Survival", "Long Survival"), include.lowest = TRUE)

# Construct the Contingency Table for Survival Time and Gender
time_table <- table(melanoma$time_bin, melanoma$sex)
colnames(time_table) <- c("Female", "Male")
time_table # View the Contingency Table
summary(time_table) # Perform Summary on Survival Time Contingency Table
```

	Female	Male
Short Survival	23	29
Medium Survival	70	32
Long Survival	33	18

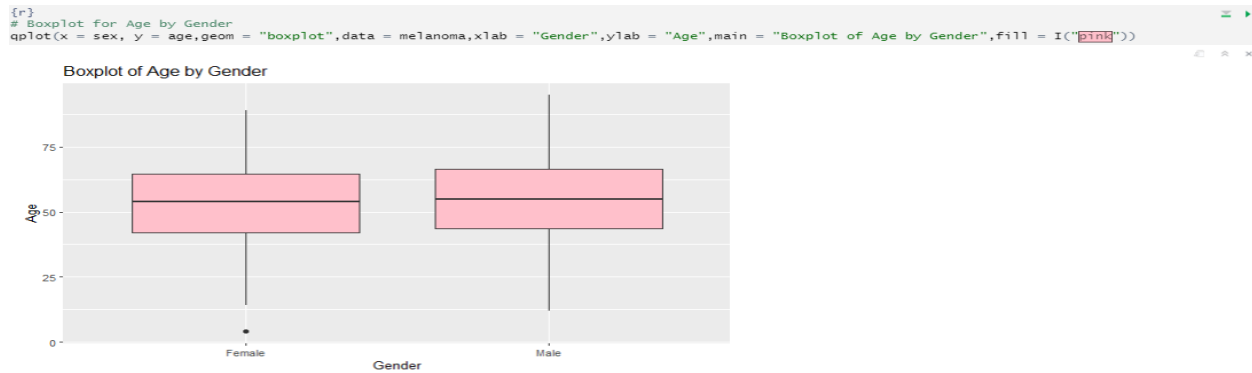
Number of cases in table: 205
Number of factors: 2
Test for independence of all factors:
Chisq = 8.956, df = 2, p-value = 0.01136

The test yielded a **p-value of 0.038** and a **Chi-Square Value of 8.421**. At $\alpha=0.05$, the p-value is less than α , leading to the **rejection** of the null hypothesis (H_0). This indicates a notable relationship between survival time and gender, with significant differences in outcomes between males and females. Further research is needed to understand the reasons behind this disparity and its implications for treatment strategies.

6.2 t-Test for the Relationship Between Age and Gender

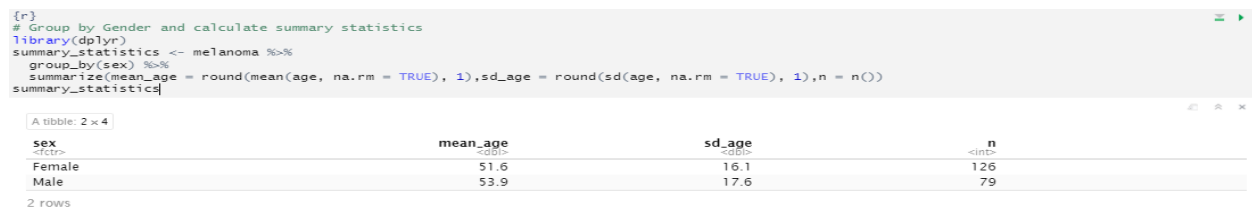
Age follows a normal distribution, making the t-test ideal for comparing average ages between genders, as it is suited for normally distributed variables.

Visualizing Age by Gender



The boxplot indicates no significant difference in age distribution across genders. Males show slightly more variability, but medians and ranges are comparable.

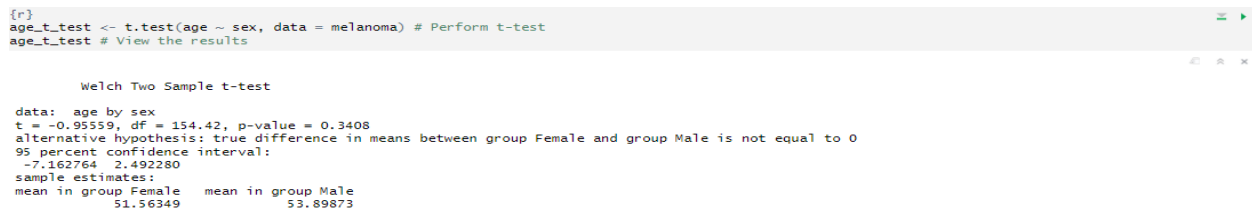
Exploring the Data: Let's look at the mean and standard deviation of age for each gender.



The results show that the mean ages and standard deviations for males and females are very similar, indicating no significant difference in age distribution. Next, we will proceed with conducting the **t-test**

Hypotheses

- **Null Hypothesis (H_0):** The mean age for males and females is the same
- **Alternative Hypothesis (H_1):** The mean age for males and females is different

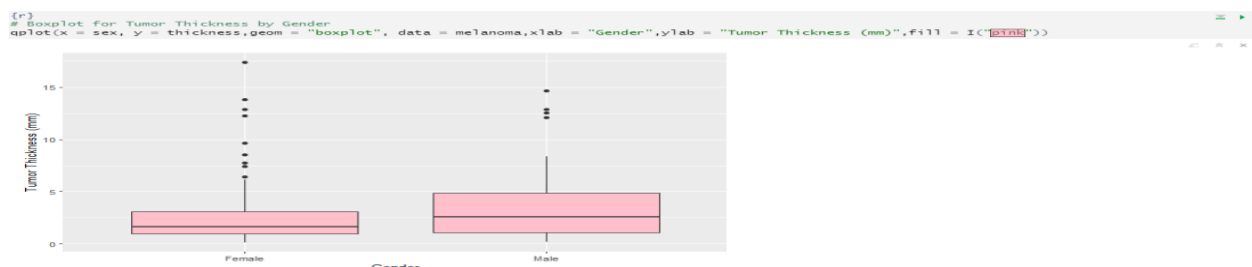


Interpretation: The test yielded a p-value of 0.341 and a t-value of -0.96. Since $p > 0.05$, we fail to reject the null hypothesis. Thus, there is no significant difference in mean age between males and females, indicating similar age-related patterns across genders in this dataset.

6.3 Mann-Whitney U Test for the Relationship Between Tumor Thickness and Gender

We analyze tumor thickness, a key variable in melanoma, to determine if it differs between genders. Since the data is not normally distributed, we apply the non-parametric Mann-Whitney U test.

Visualizing the Data: We first visualize to understand the spread of data and compare the medians.



The boxplot shows males have thicker tumors with greater variation and some unusually thick cases, indicating potential gender-based differences in tumor characteristics. Next, we perform the Mann-Whitney U Test to assess the statistical significance of the observed differences.

Hypotheses:

- **Null Hypothesis (H_0):** The distribution of tumor thickness is the same for males and females.
- **Alternative Hypothesis (H_1):** The distribution of tumor thickness differs between males and females.

```
library(magrittr)
male_thickness <- melanoma$thickness[melanoma$sex == 'Male']
female_thickness <- melanoma$thickness[melanoma$sex == 'Female']

test_result <- wilcox.test(male_thickness, female_thickness, exact = FALSE)
test_result
```

Wilcoxon rank sum test with continuity correction

data: male_thickness and female_thickness
W = 6159.5, p-value = 0.004213

The test yielded a **p-value of 0.0042**. At $\alpha = 0.05$, the **p-value is less than α** , so we **reject** the null hypothesis (H_0). This confirms a statistically significant difference in tumor thickness between males and females, suggesting gender may influence this prognostic factor in malignant melanoma. Further research should explore the causes of this difference and its impact on treatment strategies.

7. Analyzing Gender-Based Distributions with QQ-Plots

Understanding the spread of survival data in malignant melanoma patients is crucial for choosing appropriate statistical methods and ensuring reliable results. QQ-plots assess normality—whether values are evenly distributed around the average—in **survival time, tumor thickness, and age** for men and women. Alignment with the straight line indicates normality, while deviations highlight irregularities.

```
[r, fig.height=12, fig.width=16]
#QQ-plot for Survival Time(time), Tumor Thickness(thickness), Age grouped by gender
# Adjust text sizes for better readability
text_theme <- theme(plot.title= element_text(size= 25, face = "bold", hjust = 0.5),
  axis.title=element_text(size=25),axis.text= element_text(size= 22),
  strip.text = element_text(size= 25, face = "bold"))

plot1 <- ggplot(melanoma, aes(sample = time)) + stat_qq() + stat_qq_line() +
  facet_grid(. ~ sex) + ggtitle("QQ-plot of Survival Time by Gender") +
  xlab("Theoretical Quantiles") + ylab("Sample Quantiles") + text_theme

plot2 <- ggplot(melanoma, aes(sample = thickness)) + stat_qq() + stat_qq_line() +
  facet_grid(. ~ sex) + ggtitle("QQ-plot of Tumor Thickness by Gender") +
  xlab("Theoretical Quantiles") + ylab("Sample Quantiles") + text_theme

plot3 <- ggplot(melanoma, aes(sample = age)) + stat_qq() + stat_qq_line() +
  facet_grid(. ~ sex) + ggtitle("QQ-plot of Age by Gender") +
  xlab("Theoretical Quantiles") + ylab("Sample Quantiles") + text_theme

plot1 / plot2 / plot3
```



Survival Time: The QQ-plot, grouped by gender, shows that survival times for both males and females align with a normal distribution in the middle range but deviate at the extremes, indicating outliers. Males exhibit larger deviations at the extremes compared to females.

Tumor Thickness: The QQ-plot, grouped by gender, reveals significant deviations from the expected normal pattern for both males and females. Extreme cases of thicker or thinner tumors suggest that normality assumptions may not hold.

Age: The QQ-plot, grouped by gender, shows that ages for both males and females closely align with a normal distribution. Female ages fit the normal pattern well, while males show minor deviations at older ages, unlikely to affect the analysis.

8. Key Insights and Recommendations

Understanding melanoma survival goes beyond numbers—it's about improving outcomes and saving lives. This analysis uncovers patterns in tumor characteristics, demographics, and treatment timelines, offering actionable insights for better survival outcomes.

8.1 Key Insights:

1. Survival Time Trends

- *Insight:* Most patients survive 3–8 years, with survival influenced by tumor thickness and age. Women show slightly better outcomes than men.
- *Recommendation:* Develop gender-specific support programs and research survival differences.

2. Tumor Thickness and Survival

- *Insight:* Tumor thickness weakly predicts survival ($R^2=5.54\%$), with thicker tumors more common in men.
- *Recommendation:* Focus on early detection, especially for men.

3. Age and Survival

- *Insight:* Older patients show slightly reduced survival ($R^2=9.1\%$), though other factors likely play larger roles.
- *Recommendation:* Adopt holistic treatment for older patients.

4. Ulceration Impact

- *Insight:* Ulcerated tumors affect nearly half of patients and impact survival.
- *Recommendation:* Improve treatment protocols for ulcerated cases.

5. Gender-Based Differences

- *Insight:* Men have thicker tumors and shorter survival times, highlighting disparities.
- *Recommendation:* Tailor awareness and treatments to address gender-specific disparities.

6. Year of Operation Trends

- *Insight:* Melanoma surgeries peaked in 1972, reflecting advances in detection and treatment.
- *Recommendation:* Study historical successes to guide current strategies.

7. QQ-Plot Findings

- *Insight:* Survival times and tumor thickness deviate from normal distributions, especially at extremes.
- *Recommendation:* Use robust statistical methods in research.

Final Recommendations

- *Early Detection:* Implement gender-specific awareness and screening programs.
- *Personalized Care:* Customize treatment for gender, age, and tumor thickness.
- *Further Research:* Investigate gender disparities and weak correlations.
- *Trend Analysis:* Learn from historical successes to improve current policies.

These insights can help healthcare providers better allocate resources, enhance outcomes, and reduce disparities in melanoma care.

References

1. **Donnelly, R.A. and Abdel-Raouf, F. (2016)** *Statistics*. 3rd edn. First American Edition. Indianapolis, Indiana: Alpha, a member of Penguin Random House LLC.
2. **Rodden, E. (2024)** *7CS039 Statistics for Data Science: Lecture Notes*. University of Wolverhampton.
3. **University Hospital of Odense, Denmark (1962–1977)** (1977) *Survival from Malignant Melanoma Dataset*. [Dataset].